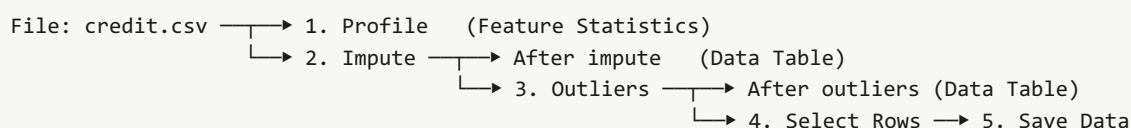


Week 2 — `week02_wrangling.ows`

Companion notes for the Orange workflow. Book: chapter 2. Notebook twin: `notebooks/week02_wrangling.ipynb`.

1. What this workflow is

A **cleaning pipeline**, left to right, with a Data Table tapped off at two points so you can see what each stage changed.



Dataset — `data/credit.csv` again (4,000 rows), because you already know from week 1 that `income` and `employment_type` have holes in them.

The idea being taught: order matters, and every step is a decision. Profiling comes first because you cannot choose an imputation strategy for a column you have not looked at. Imputation comes before outlier detection because most outlier detectors cannot handle `NaN` at all. Row filtering comes last because filtering first would change what "an outlier" means.

Note the channel names on the last three links — they are not all `Data` :

- Outliers emits `Inliers` (the rows it kept) and `Outliers` (the rows it flagged). The workflow uses `Inliers`.
- Select Rows emits `Matching Data` and `Unmatched Data`. The workflow uses `Matching Data`.

Reading channel names is how you tell what a link actually carries.

2. What to do

1. **Profile** (Feature Statistics). Confirm what you found in week 1: `income` 342 missing, `employment_type` 161.
2. **Impute**. Run it twice. First with *Average / Most frequent*, then with *Model-based imputer (simple tree)*. Open "After impute" each time and compare what landed in the `income` column.
3. **Outliers**. Switch between *One Class SVM* and *Local Outlier Factor*. Read the row count in "After outliers" each time. How many rows are dropped?
4. **Select Rows**. Add the condition `age ≥ 25`.
5. **Save Data**. Write the result to `credit_clean.tab`.
6. **Answer the canvas question:** is dropping the outliers here defensible?

Versi mahasiswa. Pembahasan, hal yang perlu diperhatikan, dan jawaban setiap langkah disampaikan di kelas setelah praktikum.